Rejection Sampling is Optimal for Relative Entropy Coding

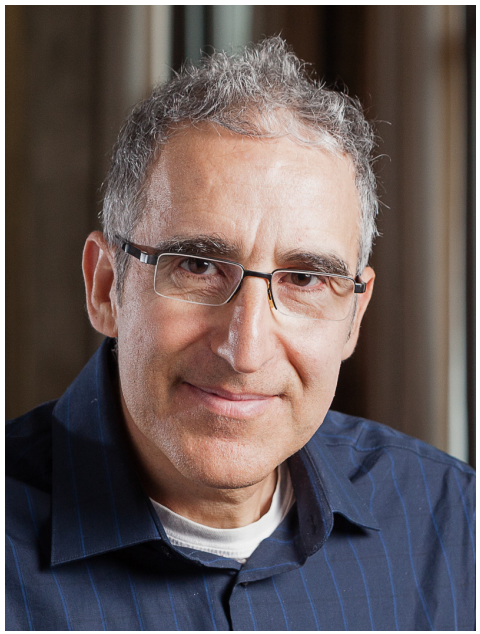
Spencer Hill

Queen's University, Canada

Hungarian Machine Learning Days

August 11, 2026

In collaboration with



Fady Alajaji

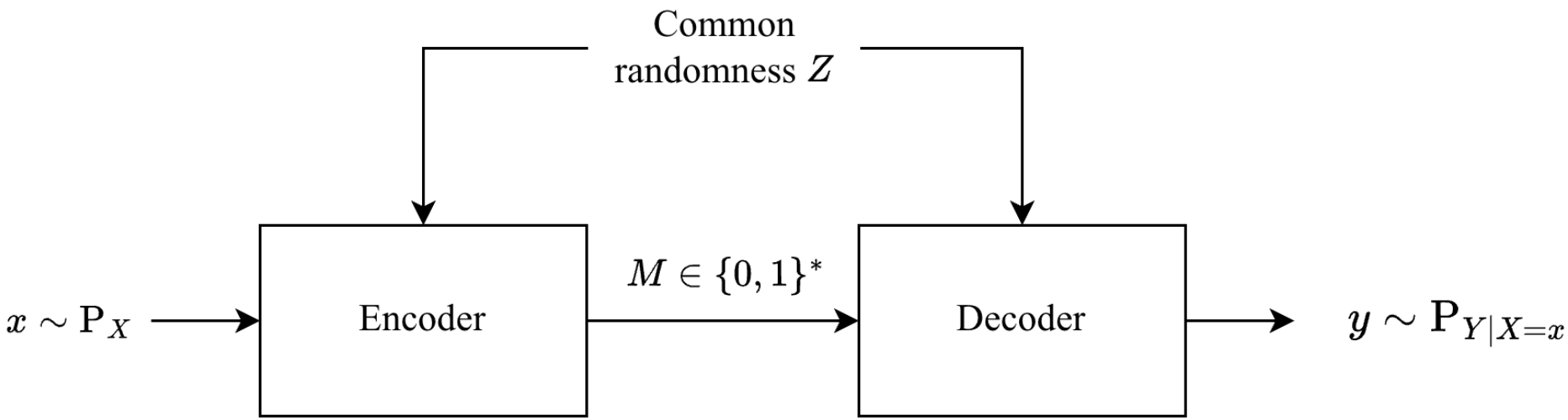


Tamás Linder



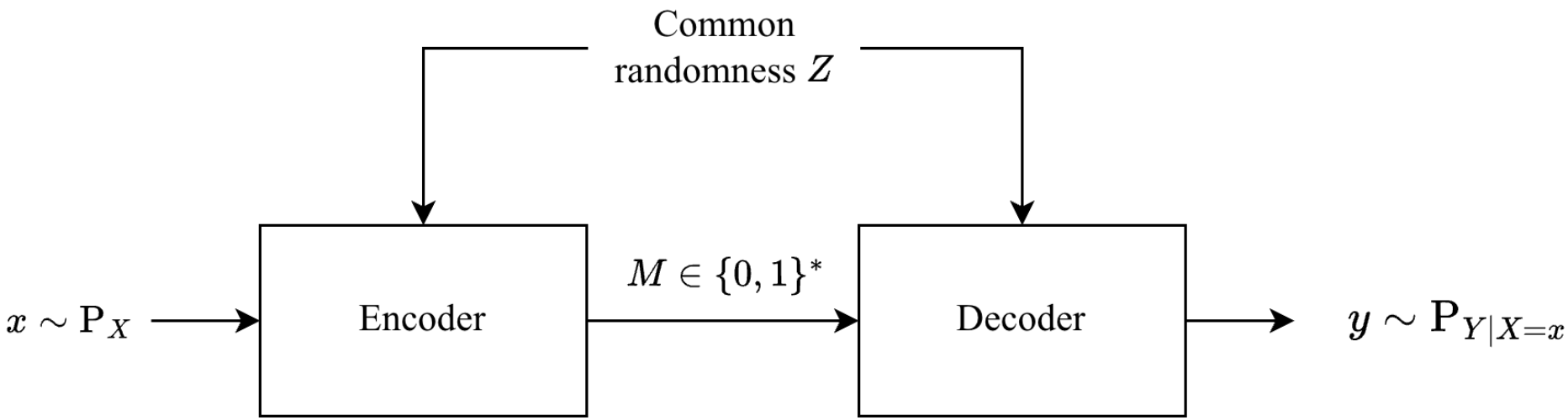
Gergely Flamich

Relative Entropy Coding



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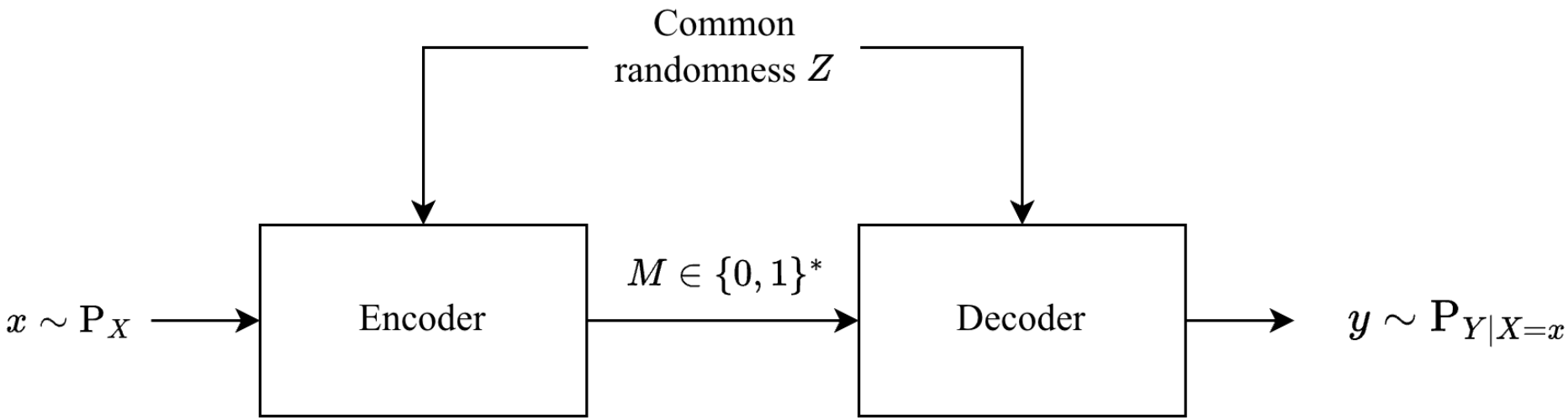
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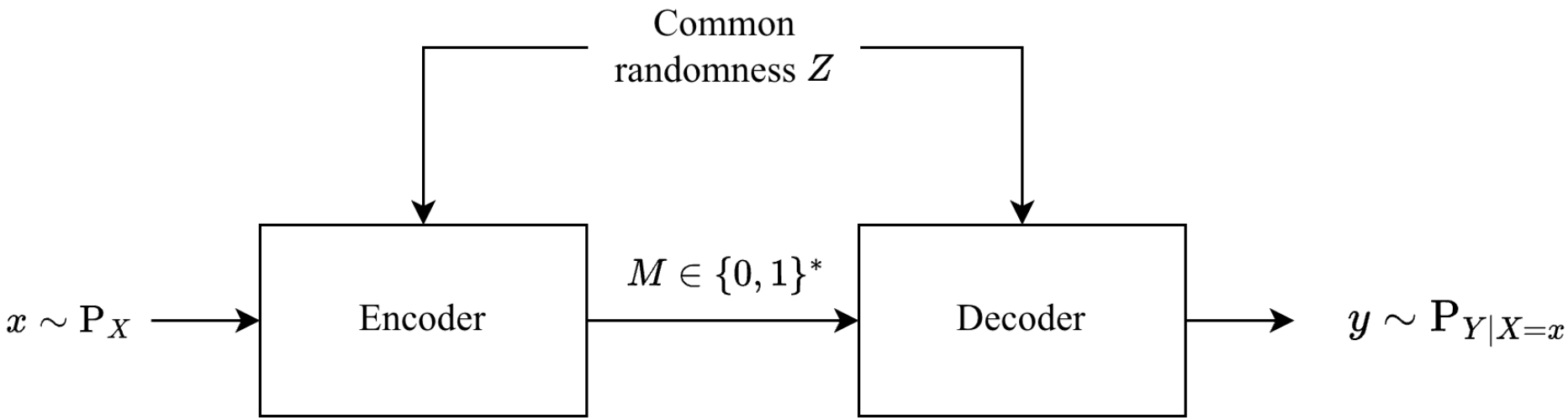
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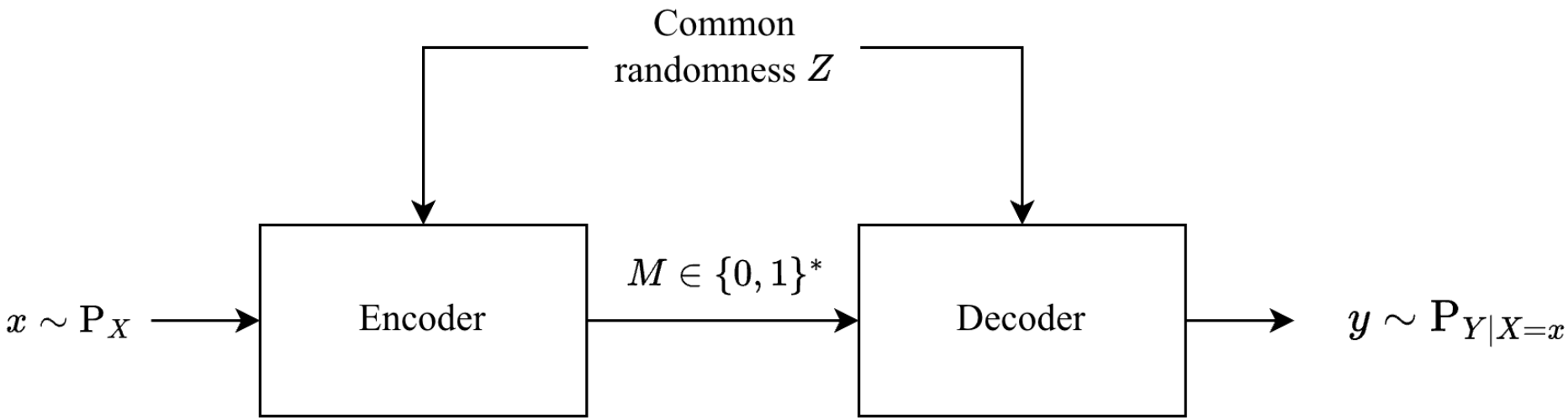
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- **Exact Requirement:** $\mathrm{dec}(\mathrm{enc}(x, Z), Z) \sim P_{Y|X=x}$ for every x

Objective and Bounds

Objective:

$$R^* := \min_{(Z,\text{enc},\text{dec})} \mathbb{E}[|\text{enc}(X,Z)|] \quad \text{s.t.} \quad \text{dec}(\text{enc}(x,Z),Z) \sim P_{Y|X=x}.$$

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This work: For $I_{\text{F}}(X \to Y)$ the *functional information*,

$$I_{\text{F}}(X \to Y) \leq R^* \leq I_{\text{F}}(X \to Y) + 2.45,$$

and the upper bound is achieved by *rejection sampling*.

Outline

- The functional information
- Rejection sampling and the ring toss code
- Application: tight one-shot rate distortion theory

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Width Function

Definition (Width Function)

For two distributions $P \ll Q$, the Q-width function at height level $\ell \geq 0$ is defined as

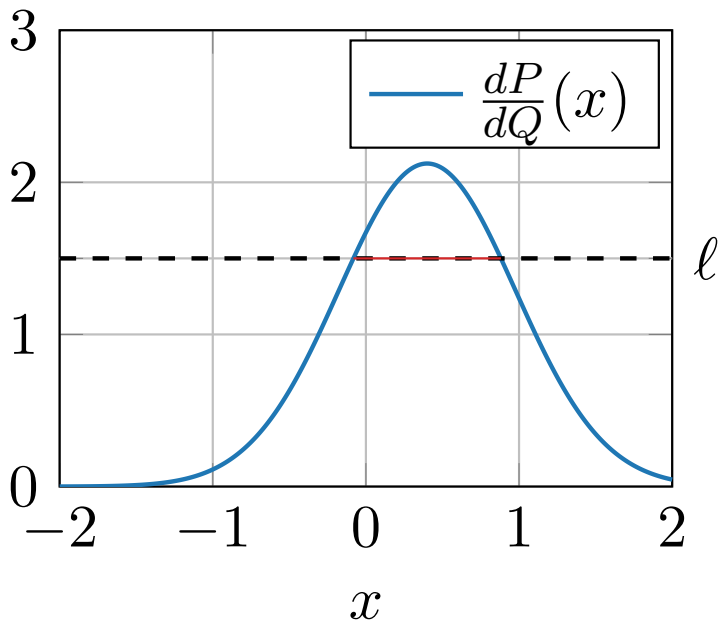
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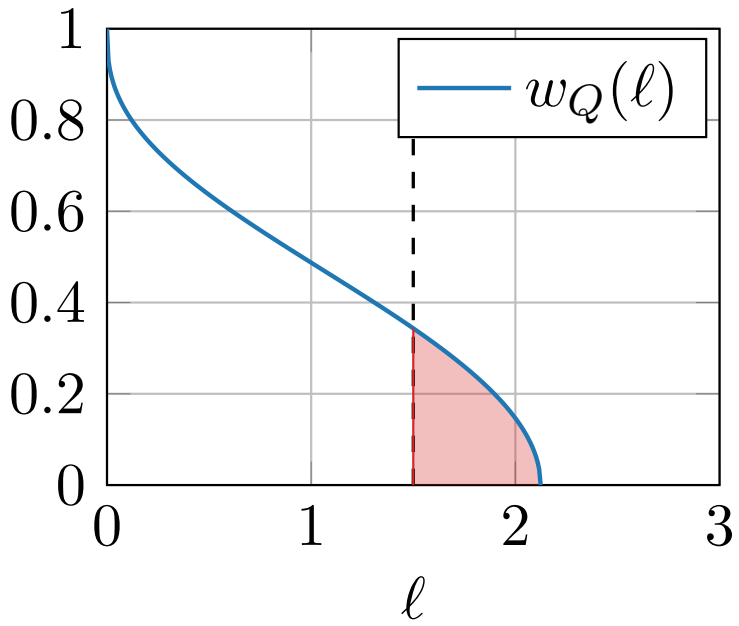
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- With $\mathrm{CS} = D_{\mathrm{CS}}(P \parallel Q)$, $\mathrm{KL} = D_{\mathrm{KL}}(P \parallel Q)$:

$$\mathrm{KL} \leq \mathrm{CS} \leq \mathrm{KL} + \log(\mathrm{KL} + 1) + 1.$$

Functional Information

Mutual Information Inspiration:

$$\begin{aligned} I(X;Y) &= D_{\mathrm{KL}}(P_{X,Y} \parallel P_X P_Y) \\ &= \mathbb{E}_X[D_{\mathrm{KL}}(P_{Y|X} \parallel P_Y)] \\ &= \mathbb{E}_Y[D_{\mathrm{KL}}(P_{X|Y} \parallel P_X)]. \end{aligned}$$

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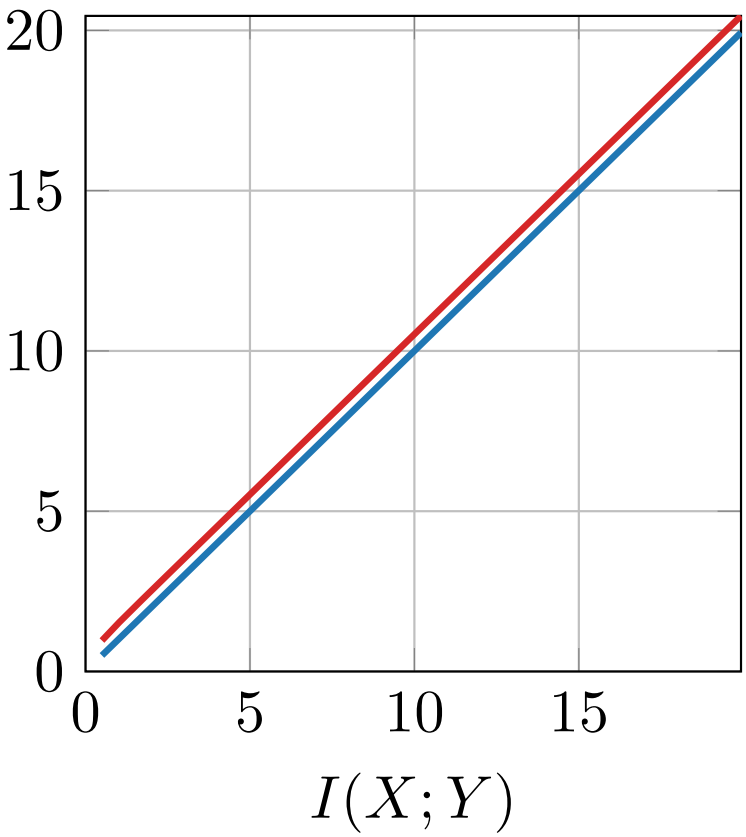
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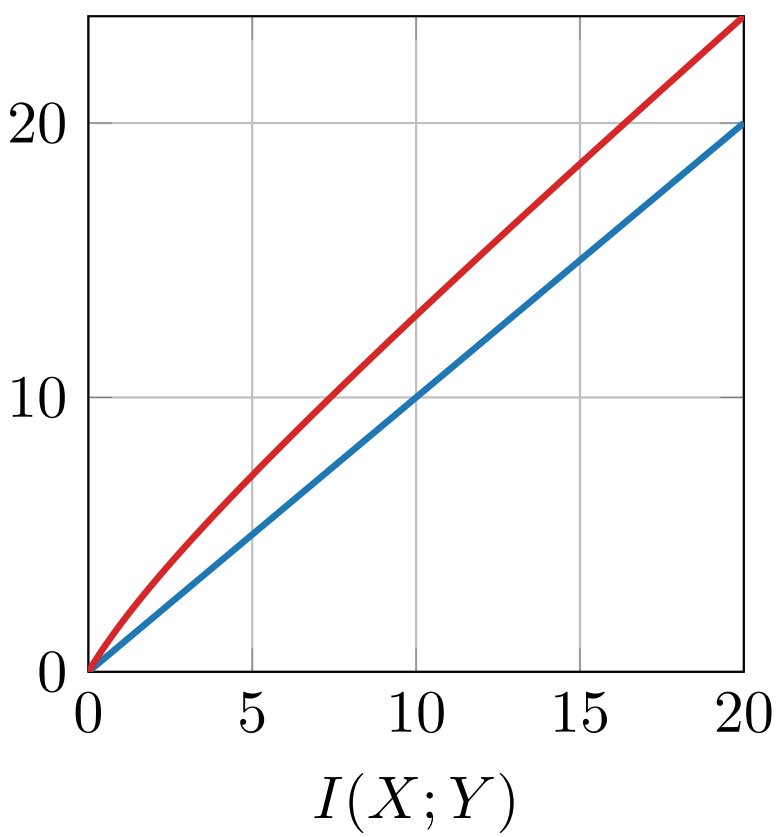
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- $I_{\mathrm{F}}(X \to Y) \leq R^*$ (best REC rate)

Some Examples

AWGN



Particular Additive Noise mod 1



$I(X;Y)$ $I_{\mathrm{F}}(X \to Y)$

Outline

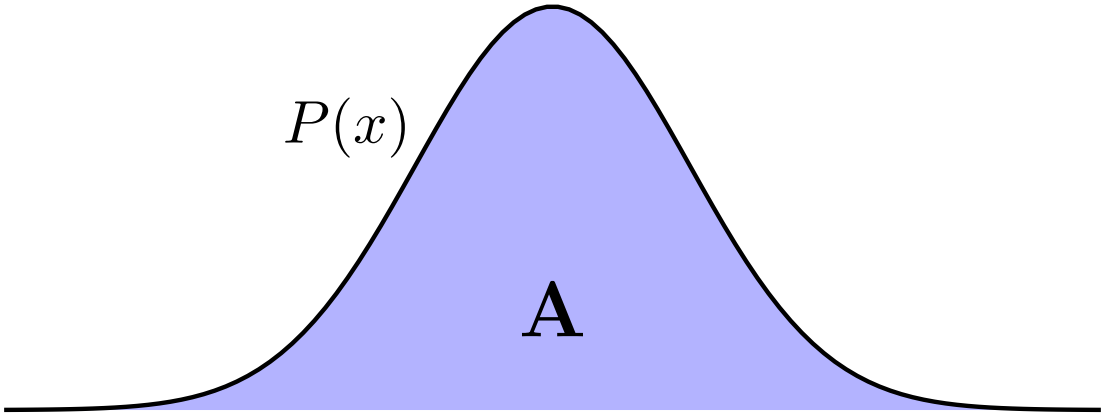
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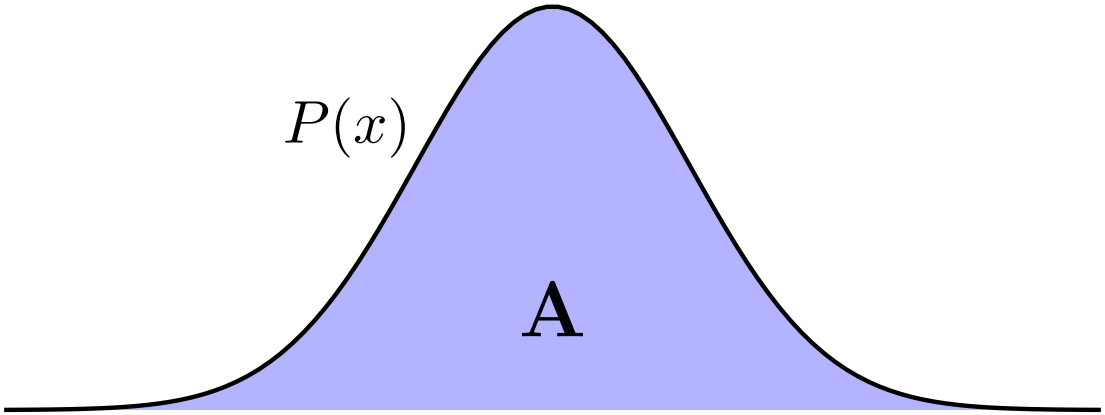
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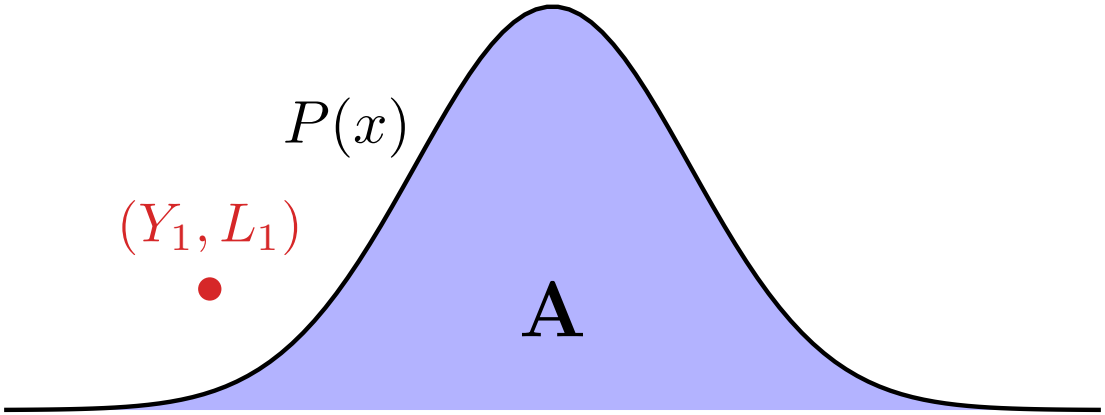
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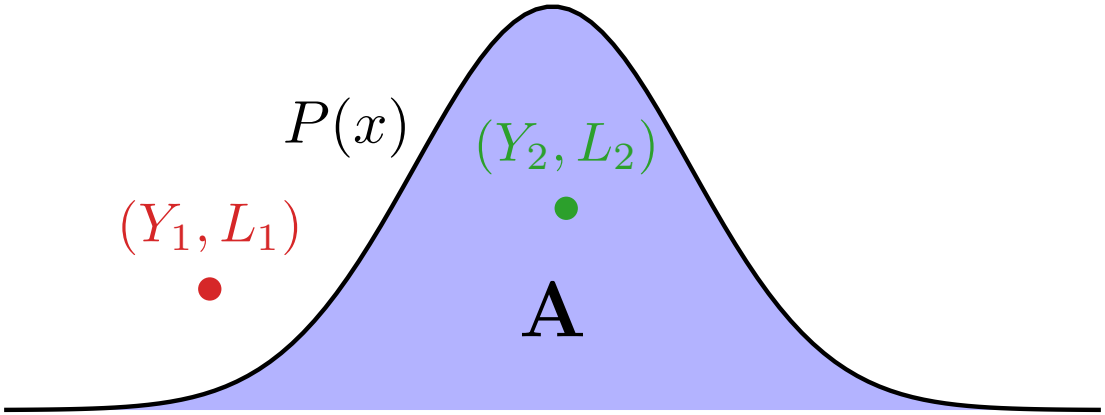
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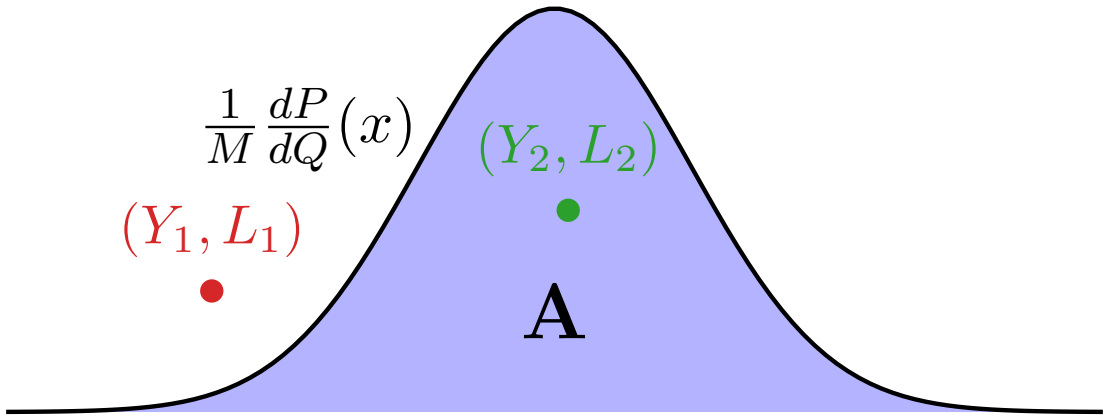
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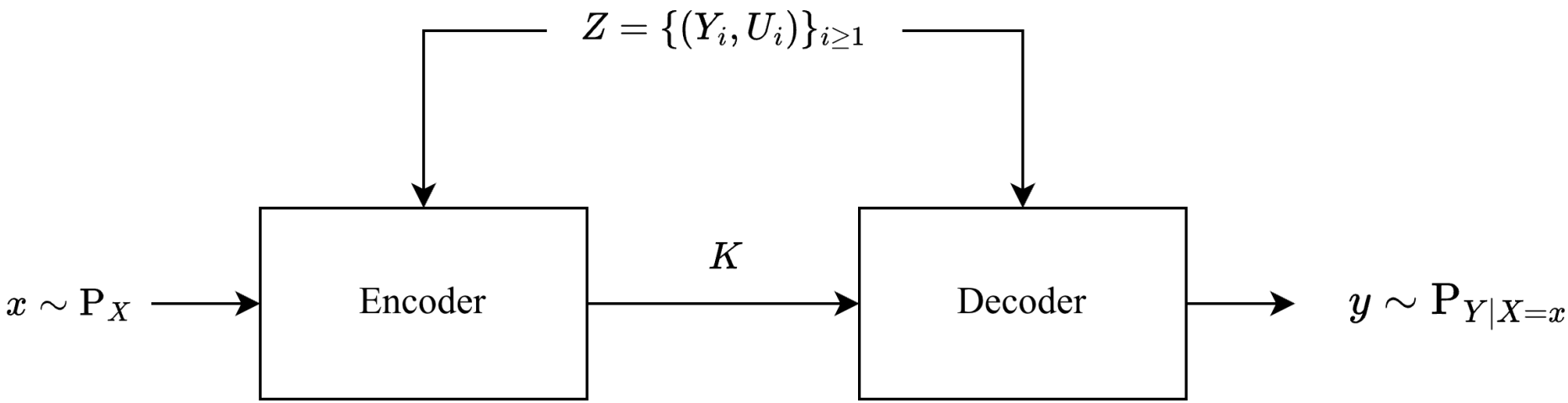
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- For general Q: accept Y_i if

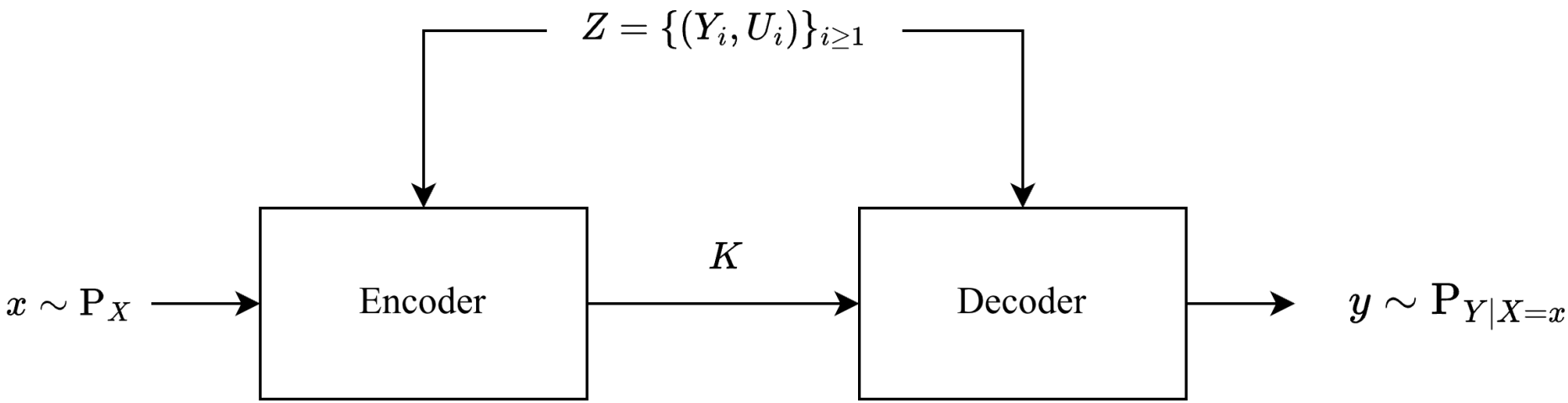
$$U_i \leq \frac{1}{M}\frac{dP}{dQ}(Y_i), \qquad \left\|\frac{dP}{dQ}\right\|_\infty \leq M.$$

Rejection Sampling as a Relative Entropy Code



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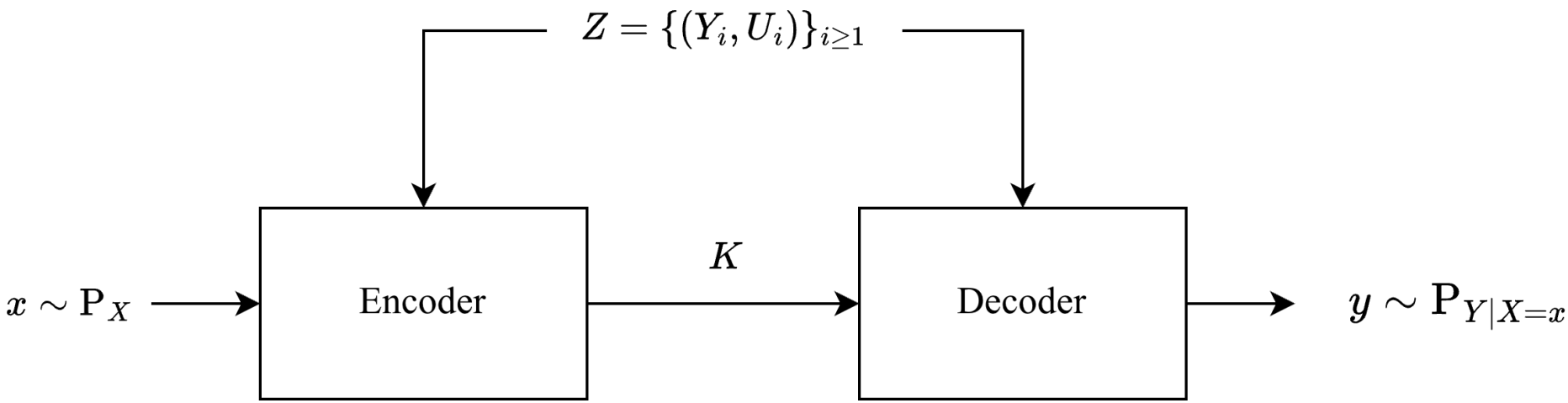
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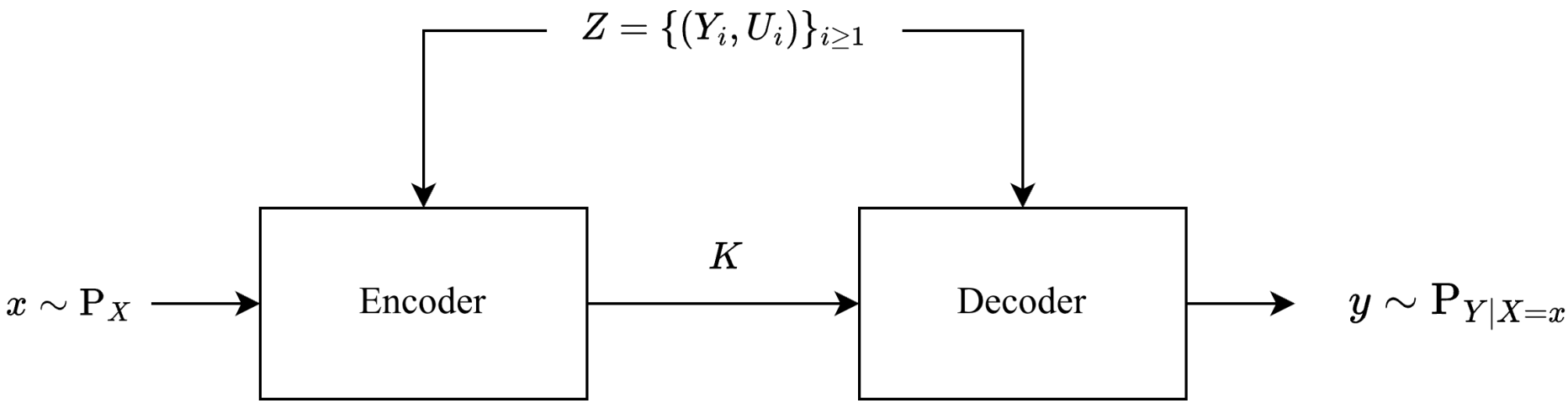
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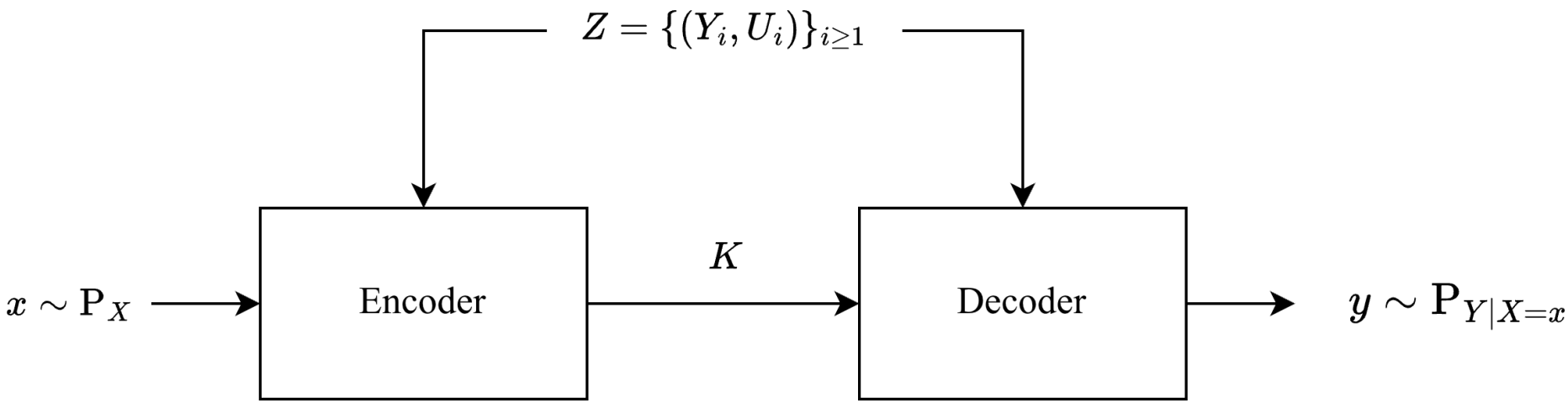
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- Send $K = \min \left\{ i : U_i \leq \frac{1}{M} \frac{dP_{Y|X}}{dP_Y}(Y_i \mid x) \right\}$

Encoding the Index K

- Unconditionally, $K \sim \text{Geom}(1/M)$

Encoding the Index K

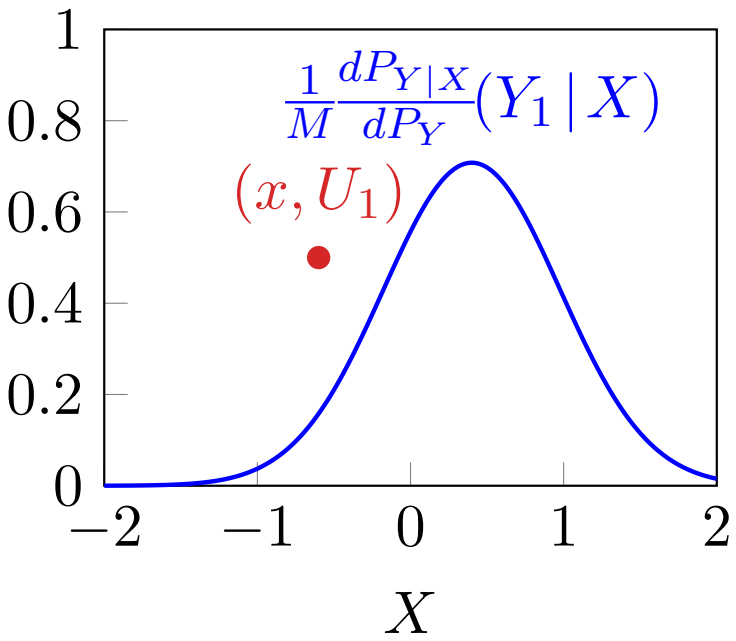
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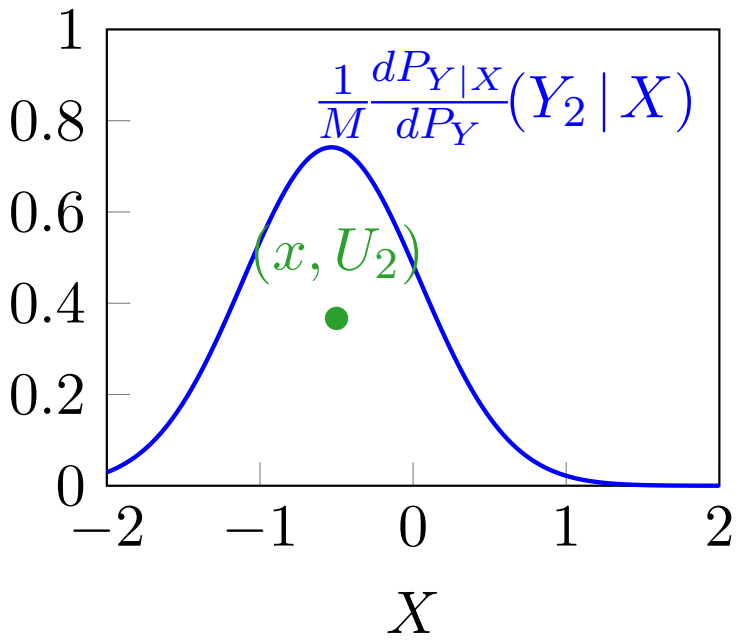
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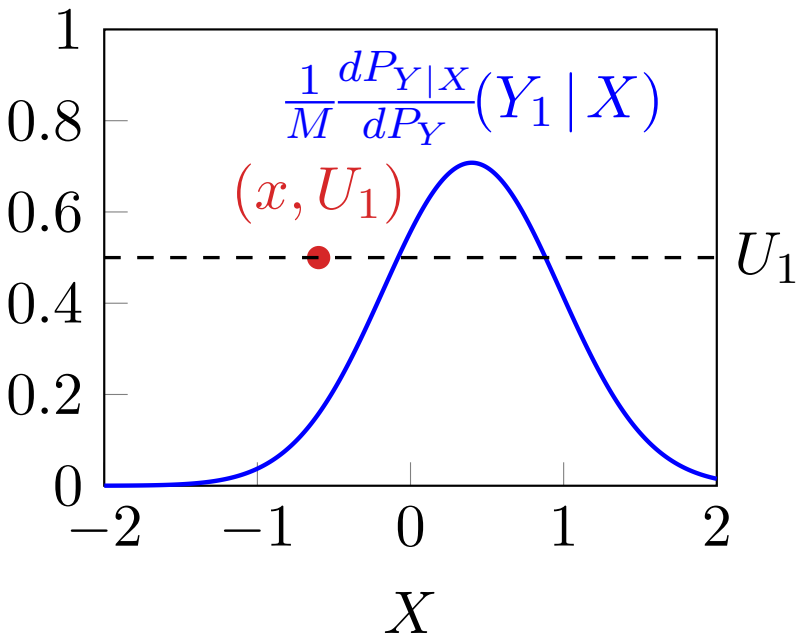
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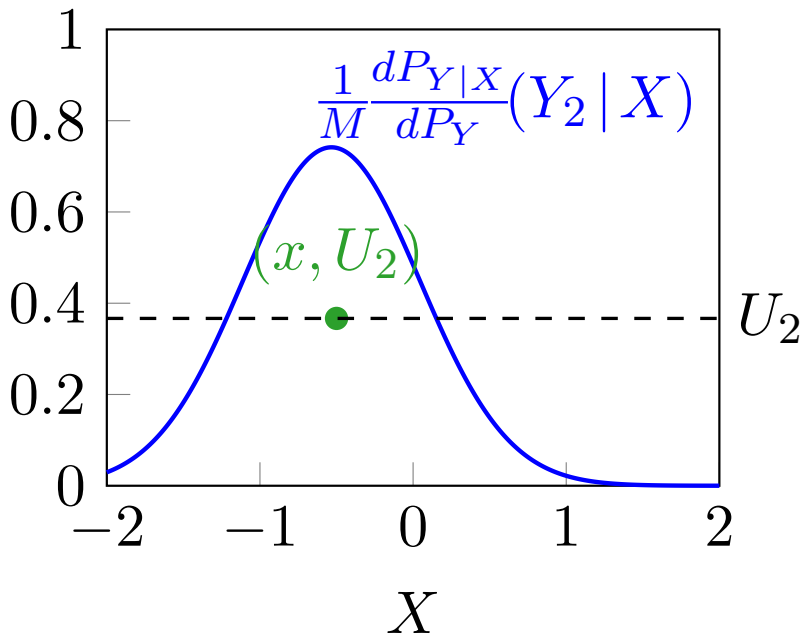




Encoding the Index K

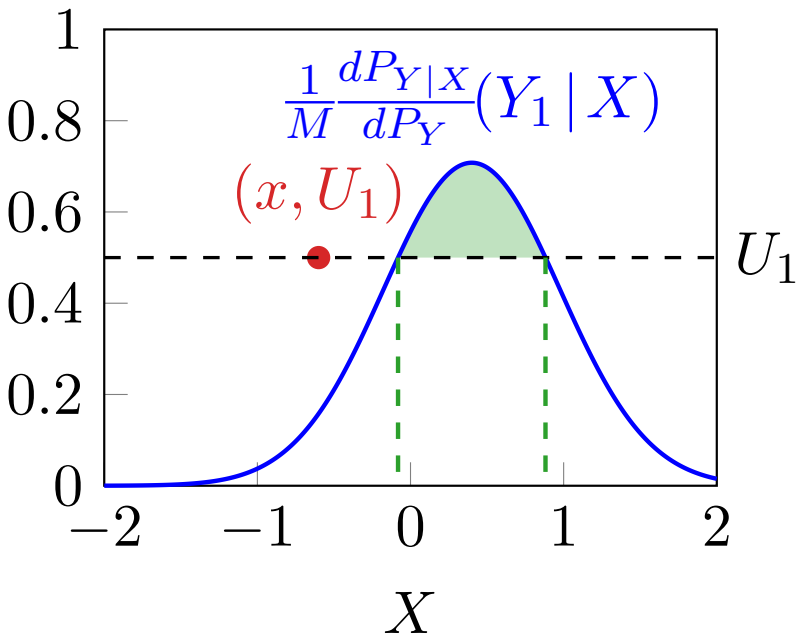
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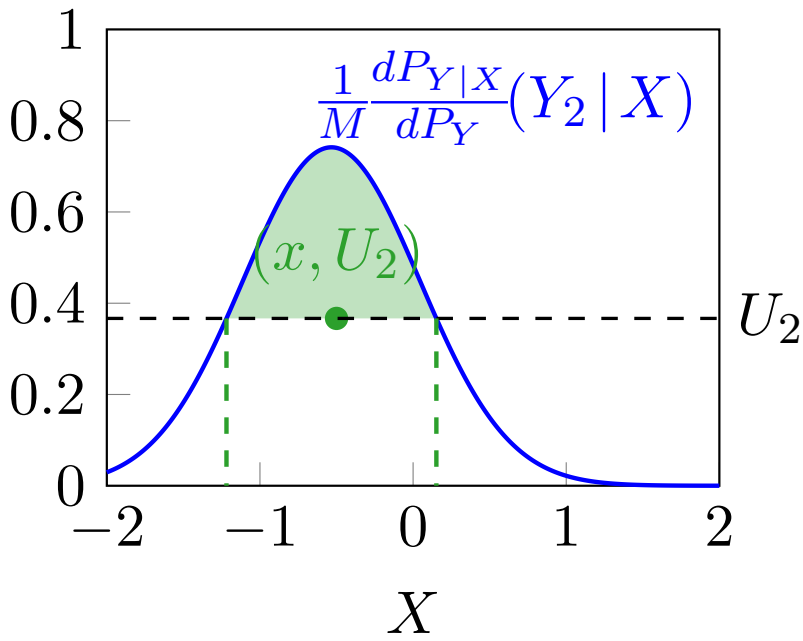




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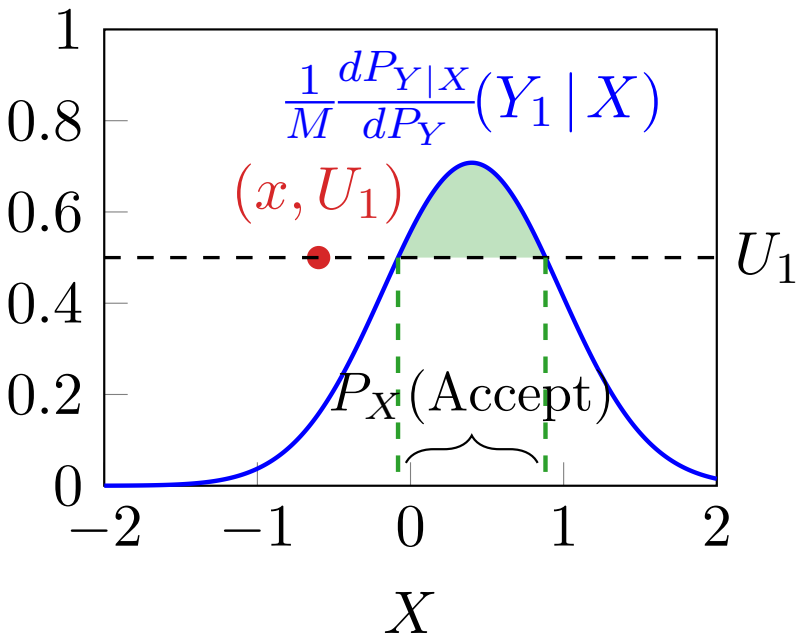
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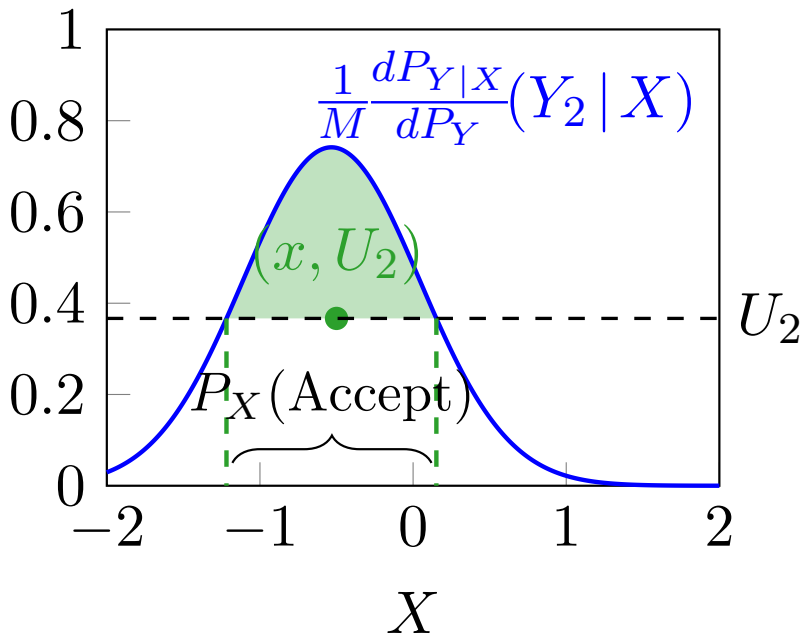




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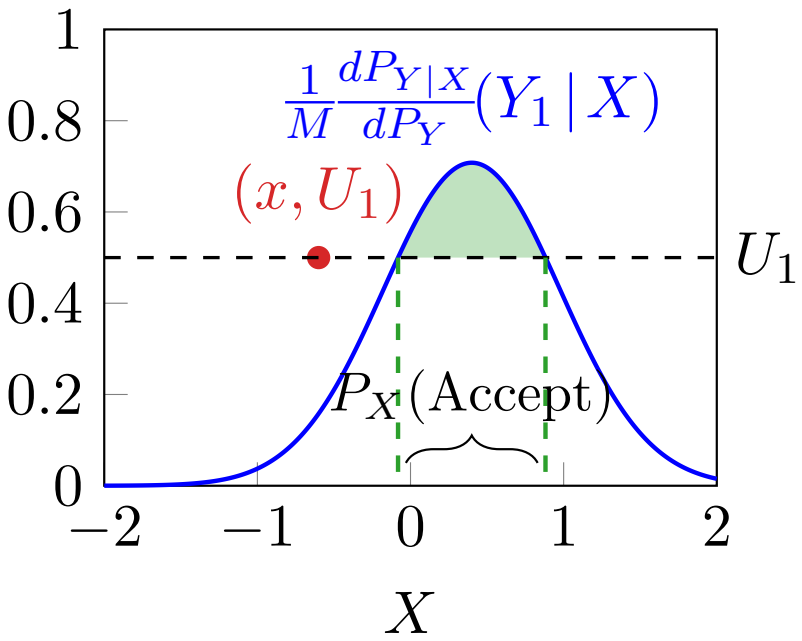
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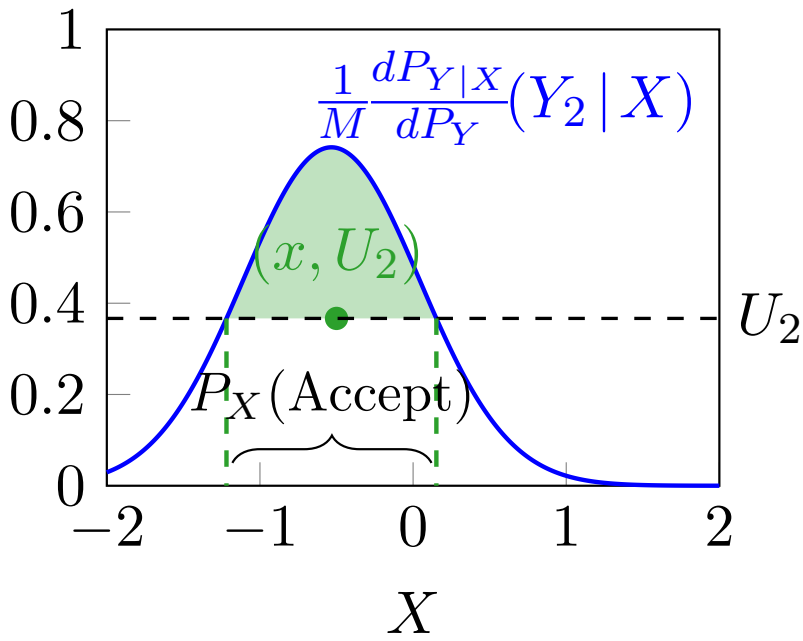




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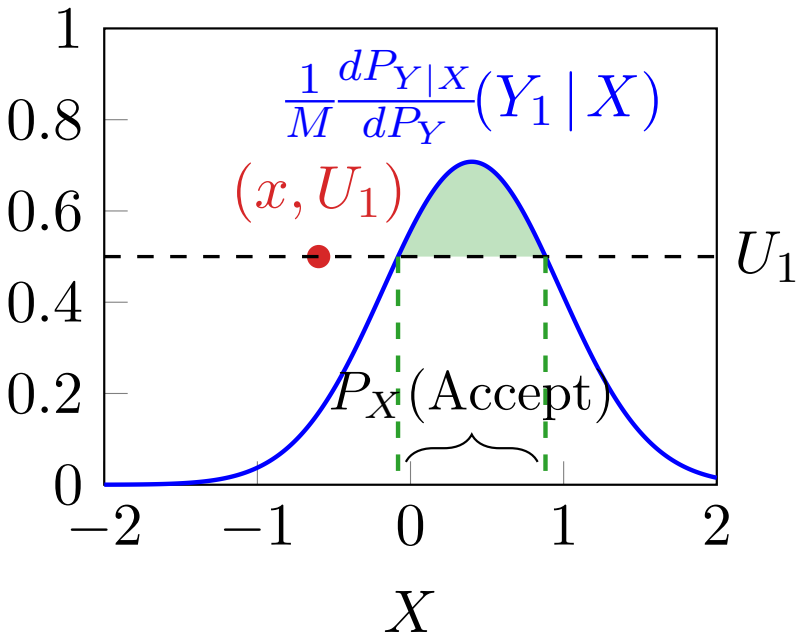


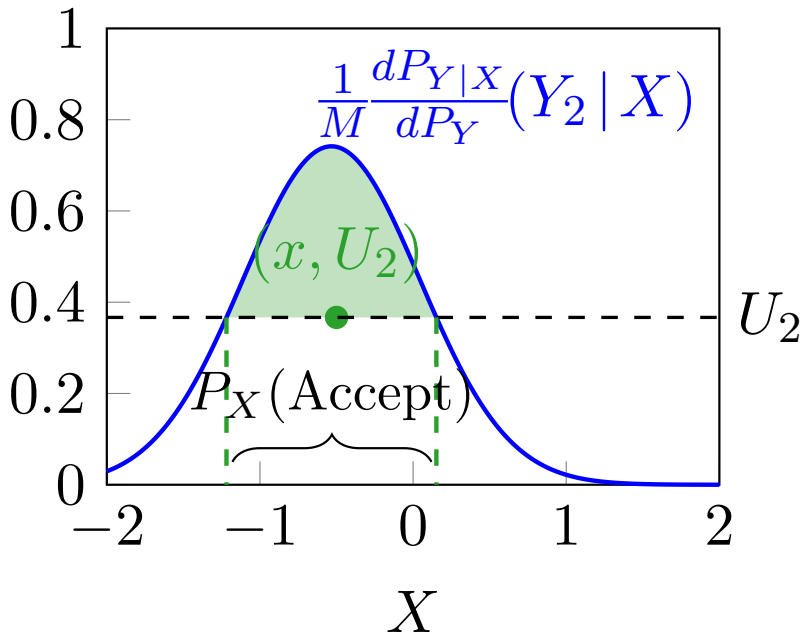


$P_X(\text{Accept } i)$

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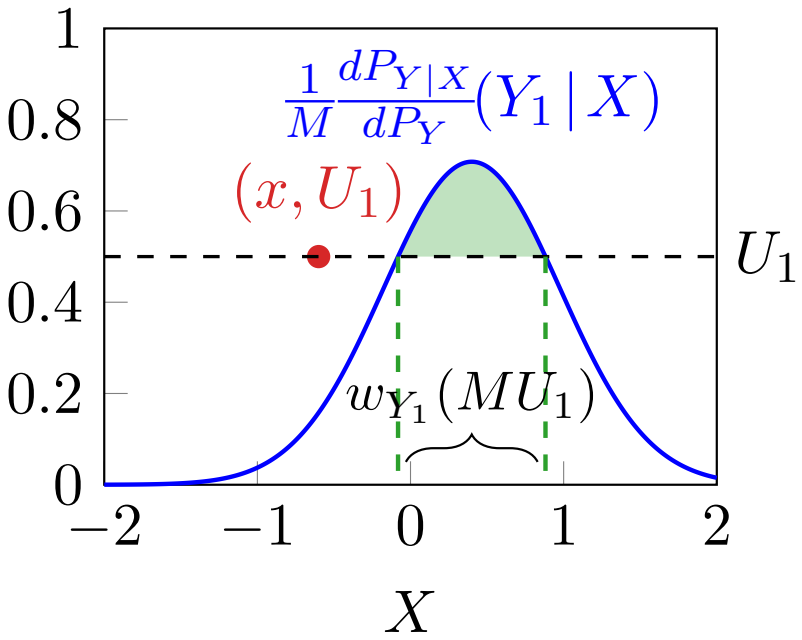


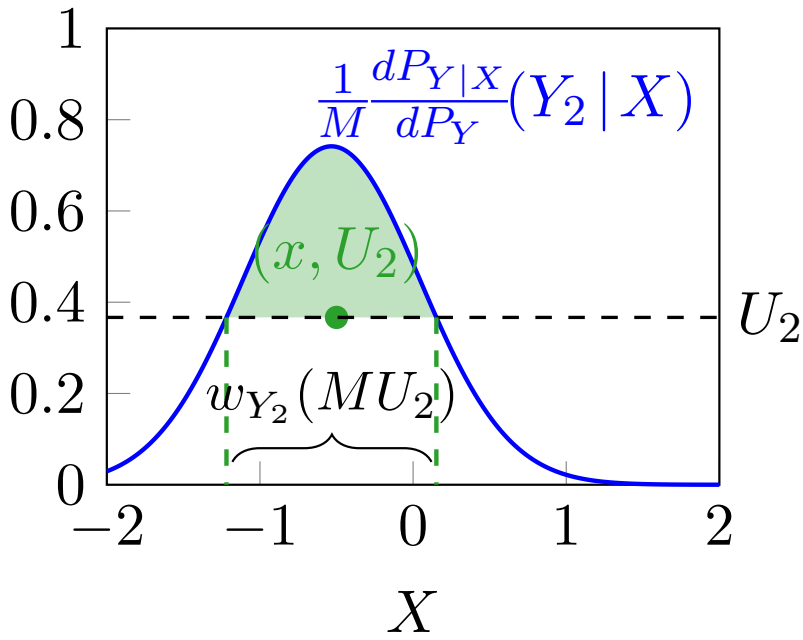


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$$P_X(\text{Accept } i) = P_X\left(\frac{dP_{Y|X}}{dP_X}(Y_i \mid X) \geq MU_i\right) =: w_{Y_i}(MU_i)$$

Ring Toss Code

Encode K using a (prefix-free) code optimal for the distribution

$$Q_{K|Z}(k \mid z) = w_{y_k}(Mu_k) \prod_{i=1}^{k-1} (1 - w_{y_i}(Mu_i)).$$

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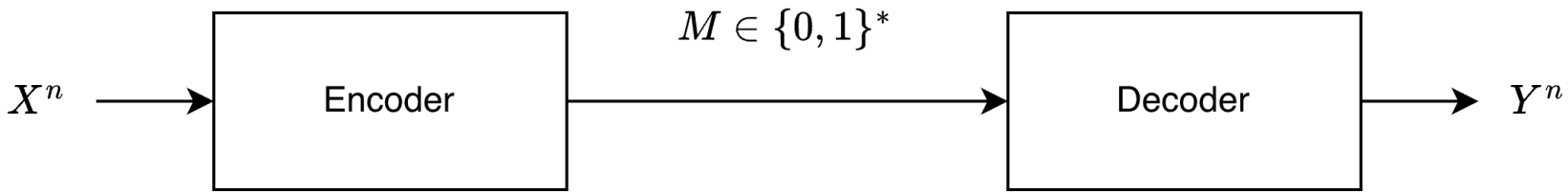
- **Tight characterization:** $I_{\mathrm{F}}(X \to Y) \leq R^* \leq I_{\mathrm{F}}(X \to Y) + 2.45$
- **Improved mutual information bound:**

$$I(X;Y) \leq R^* \leq I(X;Y) + \log(I(X;Y) + 1) + 3.45.$$

Outline

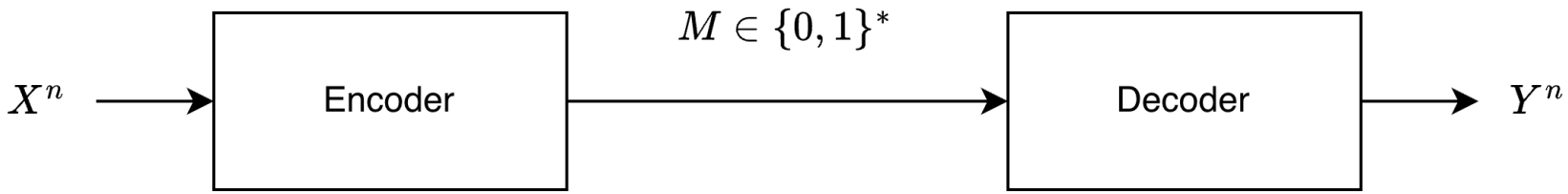
- The functional information
- Rejection sampling and the ring toss code
- Application: tight one-shot rate distortion theory

Lossy Data Compression



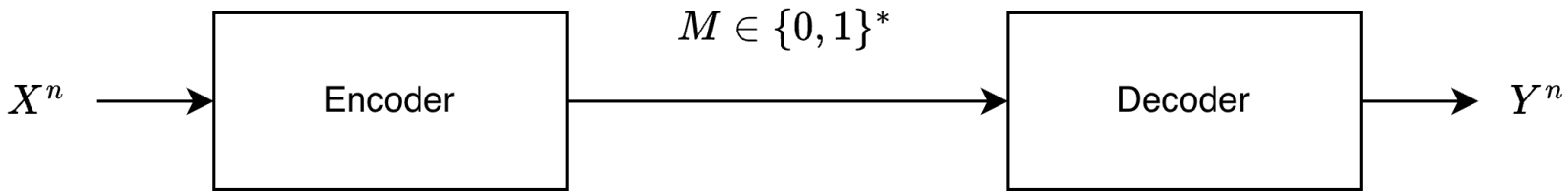
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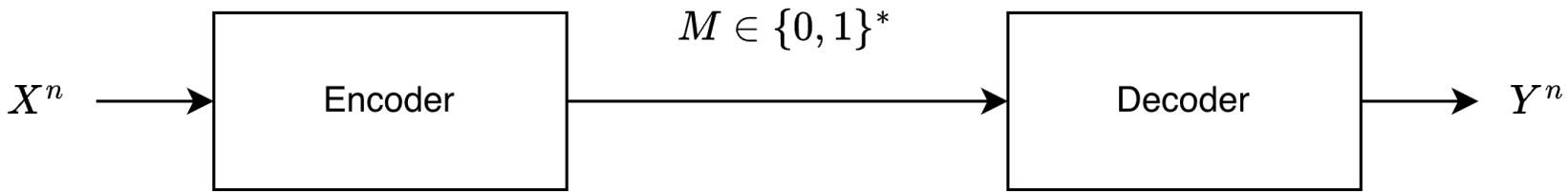
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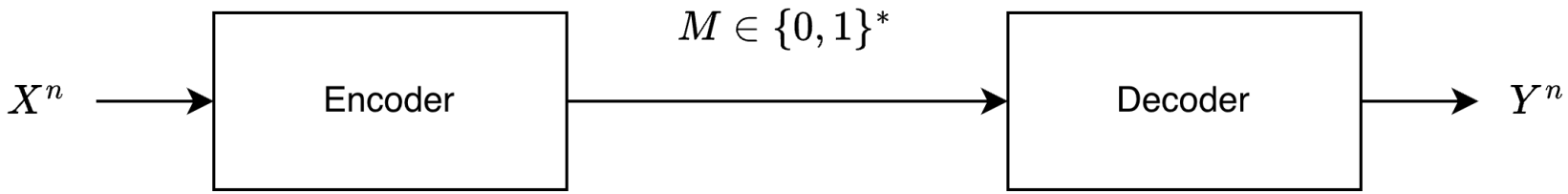
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- Rate: $R = \frac{1}{n}\mathbb{E}[|\text{enc}(X^n)|]$ (expected message length)
- Shannon (1948): *Asymptotically* ($n \to \infty$) optimal performance:

$$R(D) = \min_{\mathrm{P}_{Y|X} \,:\, \mathbb{E}[\Delta(X,Y)] \leq D} I(X;Y).$$

One-Shot Rate Distortion Theory

One-shot rate distortion function:

$$R_O(D) := \inf_{(\text{enc},\text{dec})} \mathbb{E}[|\text{enc}(X)|] \quad \text{s.t.} \quad \mathbb{E}[\Delta(X,Y)] \leq D.$$

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Theorem

For any $D > 0$,

$$R_F(D) \leq R_O(D) \leq R_F(D) + 3.45,$$

where

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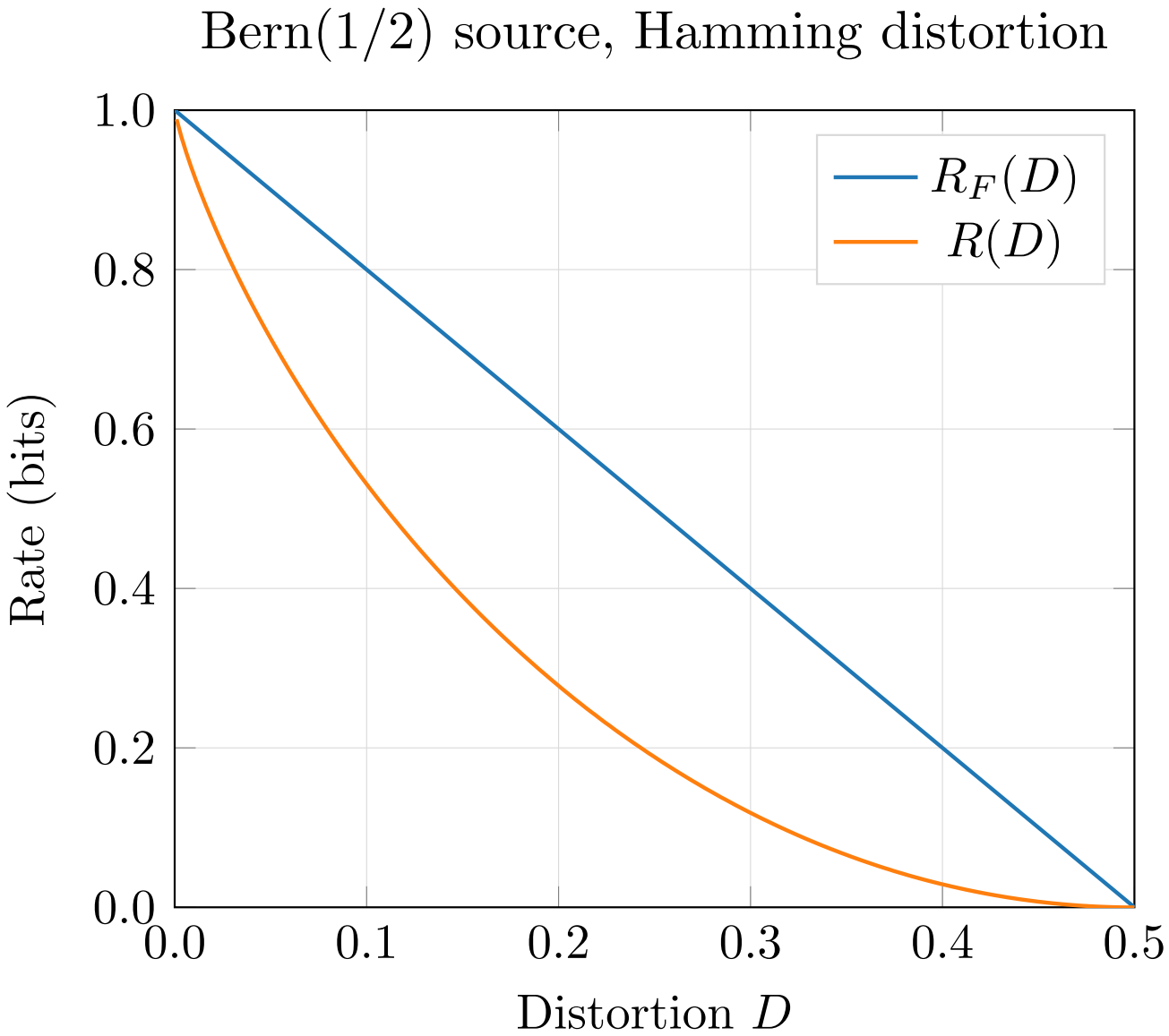
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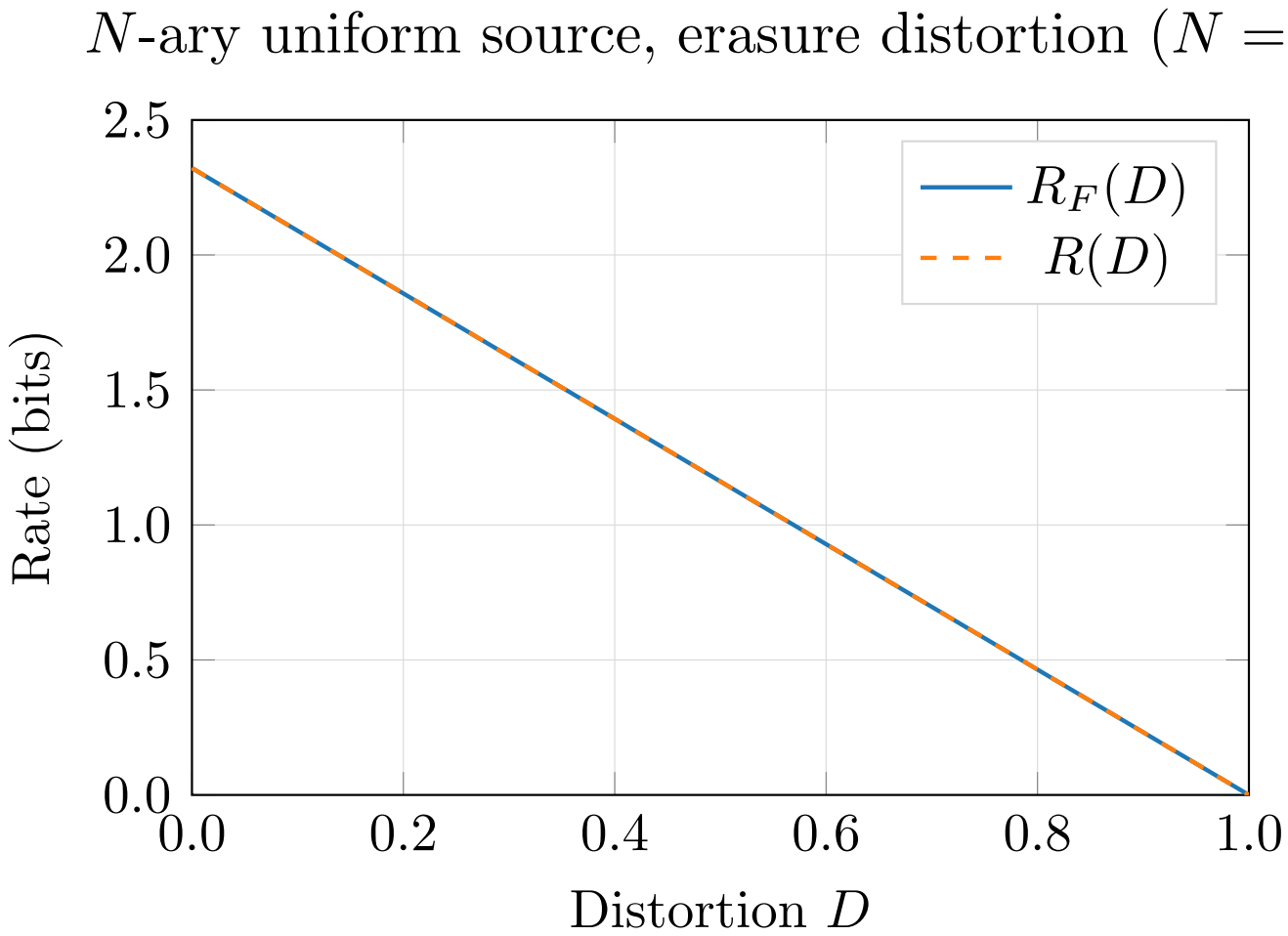
Similarly, for $R_{O,n}(D) \coloneqq \frac{1}{n} R_O(nD)$ and $R_{F,n}(D) \coloneqq \frac{1}{n} R_F(nD)$,

$$R_{F,n}(D) \leq R_{O,n}(D) \leq R_{F,n}(D) + \frac{\log e + 2}{n}.$$

Examples


Bern(1/2) source, Hamming distortion
1.0
0.8
0.6
0.4
0.2
0.0
Rate (bits)
0.0
0.1
0.2
0.3
0.4
0.5
Distortion D
$R_F(D)$
$R(D)$

Examples



The minimizing channel $X \to Y$ is singular, so $R(D) = R_F(D)$ is achievable using a one-shot code!

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Read the arXiv paper!

